Telecom Customer Churn Prediction & Retention Analytics

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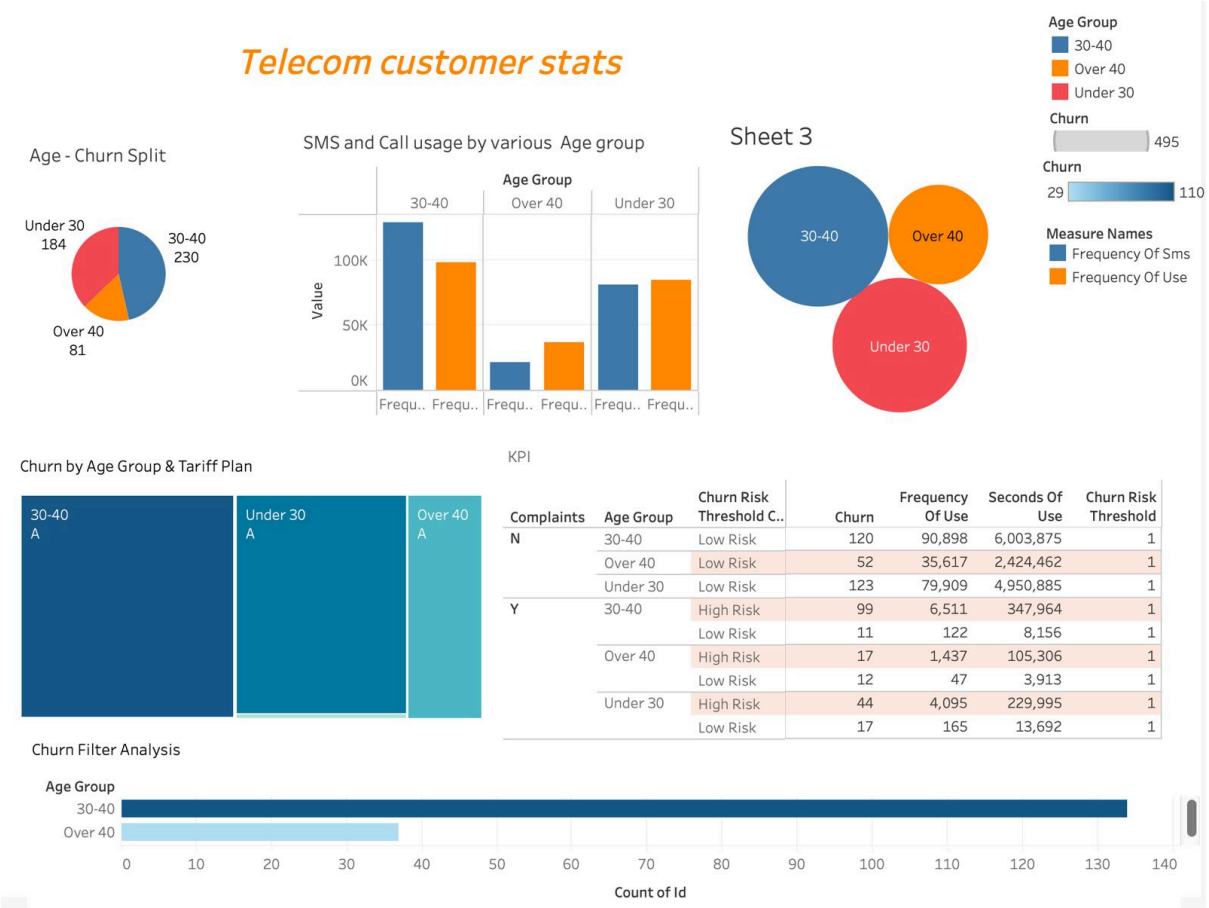
1. Executive Summary

Customer churn is one of the most significant challenges faced by telecom companies, as it directly affects recurring revenue, profitability, and long-term customer relationships. In a highly competitive market where switching costs are low, retaining existing customers becomes more critical than acquiring new ones.

This project focuses on identifying customers who are likely to churn in the upcoming quarter using historical telecom data. By analyzing customer usage behavior, service quality indicators, and engagement patterns, the project aims to uncover actionable insights that can help businesses proactively reduce churn.

A structured approach was followed, beginning with data extraction and cleaning, followed by exploratory data analysis (EDA), KPI framework development, and preparation for statistical modeling. The outcomes of this project provide both strategic and operational insights for improving customer retention.



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2. Sector Context

The telecom industry operates in a highly saturated and competitive environment. Customers have multiple service providers to choose from, often with similar pricing and service offerings. This leads to a scenario where customer loyalty is low and churn rates are high.

Key industry challenges include:

- Increasing customer churn due to competitive offers
- Poor service quality leading to dissatisfaction
- Lack of personalized customer engagement strategies
- Inefficient identification of high-risk customers

In this context, data analytics plays a crucial role in understanding customer behavior and predicting churn. By leveraging historical data, telecom companies can move from reactive problem-solving to proactive decision-making.

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3. Problem Statement

The primary objective of this project is to identify customers who are likely to churn in the next quarter and to determine the key factors influencing their decision to leave the service.

The problem requires not only identifying churn patterns but also translating these patterns into actionable business insights that can guide retention strategies.

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4. Objectives

The project is guided by the following objectives:

- To analyze customer behavior using usage and engagement data
- To identify key drivers contributing to customer churn
- To develop a KPI framework that reflects customer health and risk
- To segment customers based on churn probability
- To prepare a clean and structured dataset for statistical modeling
- To generate actionable insights for business decision-making

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5. Data Description

The dataset used in this project represents telecom customer activity and includes multiple dimensions of customer interaction. It contains approximately 3,150 records and 26 features after processing.

Key categories of features include:

- **Subscription Information:** subscription length, account details
- **Usage Metrics:** seconds of use, frequency of calls and SMS
- **Engagement Indicators:** frequency of interactions with services
- **Service Quality Metrics:** call failures and network-related issues
- **Billing Information:** charge amounts and usage-related costs
- **Target Variable:** churn indicator representing whether a customer has left the service

This dataset provides a comprehensive view of customer behavior, making it suitable for churn analysis.

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6. Data Cleaning & ETL Pipeline

A robust data engineering pipeline was implemented to ensure that the dataset is clean, consistent, and analysis-ready.

Key Steps Performed:

- **Handling Missing Values:**
  Numerical columns were filled using median values, while categorical columns were filled using the most appropriate category or labeled as "Unknown" where necessary.
- **Duplicate Removal:**
  Duplicate records were identified and removed to maintain data integrity.
- **Data Type Corrections:**
  Columns were converted into appropriate formats (e.g., numeric, categorical) to ensure correct analysis.
- **Column Standardization:**
  All column names were converted to snake_case for consistency and readability.
- **Outlier Detection and Treatment:**
  Outliers were identified using statistical techniques such as the IQR method and capped where necessary to prevent distortion in analysis.
- **Data Consistency Checks:**
  Inconsistent labels and formatting issues were corrected.

The final cleaned dataset was stored as:

data/processed/cleaned_data.csv

All transformations were documented in detail to maintain transparency and reproducibility.

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7. KPI & Metric Framework

To align the analysis with business objectives, a structured KPI framework was developed. These KPIs provide measurable indicators of customer behavior and churn risk.

KPI	Definition	Business Interpretation
Churn Rate	Percentage of customers who churn	Measures overall retention performance
Usage Intensity	Usage divided by subscription length	Indicates how actively a customer uses the service
Engagement Intensity	Interaction frequency over time	Reflects customer involvement
SMS Intensity	SMS usage behavior	Shows communication preference
Call Failure Rate	Failed calls relative to usage	Measures service quality issues

These KPIs were used throughout the analysis to ensure consistency and business relevance.

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8. Exploratory Data Analysis (EDA)

EDA was conducted to understand patterns, relationships, and trends in the dataset. The analysis focused on identifying key differences between churned and retained customers.

8.1 Univariate Analysis

Individual variables were analyzed to understand their distributions. This helped identify skewness, outliers, and unusual patterns in customer behavior.

8.2 Bivariate Analysis

Key variables were compared against the churn indicator to identify relationships.

This analysis revealed how customer behavior differs between churned and non-churned groups.

8.3 Segment Analysis

Customers were divided into segments based on usage, engagement, and service quality.

This segmentation helped identify high-risk groups and understand behavioral differences across segments.

8.4 Correlation Analysis

A correlation matrix was used to identify relationships between numerical variables.

This helped detect redundant features and strong predictors of churn.

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9. Key Insights

The analysis generated several important insights:

- Customers with low engagement intensity are significantly more likely to churn
- High call failure rates are strongly associated with churn
- Customers with shorter subscription durations have higher churn risk
- Low-usage customers represent a critical high-risk segment

- Customers with minimal charges show higher churn probability
- Service quality issues directly influence customer retention
- Behavioral metrics are stronger predictors than static attributes
- Certain segments contribute disproportionately to churn
- Early-stage customers require targeted retention strategies
- Normalized KPIs provide better insights than raw values
- Engagement patterns play a crucial role in customer retention
- Churn is influenced by a combination of multiple behavioral factors rather than a single variable

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10. Statistical Analysis Readiness

The dataset was further prepared for statistical analysis and modeling.

Key Enhancements:

- Creation of derived KPI features
- Removal of missing values
- Encoding of categorical variables
- Standardization of numerical fields

Data Separation Strategy:

To ensure modeling accuracy and prevent data leakage:

- **final_kpi_data.csv** was used for dashboarding
- **final_model_data.csv** was created for modeling after removing leakage-prone features such as status and complaints

This separation ensures that predictive models are unbiased and reliable.

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11. Tableau Dashboard Design

An interactive Tableau dashboard was developed to visualize key insights and KPIs.

Dashboard Features:

- Overview of churn rate and customer distribution
- Segmentation of customers based on behavior
- Visualization of usage, engagement, and service quality
- Interactive filters for deeper analysis

The dashboard enables stakeholders to explore data dynamically and make informed decisions.

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12. Recommendations

Based on the insights derived from the analysis, the following recommendations are proposed:

1. Implement targeted engagement campaigns for low-activity customers
2. Improve network reliability to reduce call failures
3. Focus on retaining customers in the early stages of subscription
4. Offer incentives to increase usage among low-engagement users
5. Monitor key KPIs regularly to detect early signs of churn

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13. Business Impact

If implemented effectively, these recommendations can lead to:

- Reduction in churn rates
- Increased customer lifetime value
- Improved customer satisfaction
- More efficient allocation of retention resources

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14. Limitations

- The dataset may not include all external factors influencing churn
- Some variables may act as lagging indicators
- Limited time-based data restricts long-term trend analysis

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15. Future Scope

Future improvements can include:

- Building predictive machine learning models
- Integrating real-time data for live churn prediction
- Expanding the dataset with additional customer interaction data
- Enhancing dashboards with advanced analytics features

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16. Conclusion

This project provides a comprehensive and structured approach to analyzing customer churn in the telecom sector. By combining data engineering, exploratory analysis, KPI development, and statistical readiness, the project delivers meaningful insights that can drive business decisions.

The approach enables organizations to move toward proactive customer retention strategies, ultimately improving long-term profitability and customer satisfaction.